Advanced Machine Learning, ETH Zurich

U-Nets

Carlos Cotrini

Autumn semester 2026

A U-Net turns an image into a segmentation map

Image segmentation: for every pixel of an image, find the object it shows. Our example has two objects, a dog and the background.



input X: $h \times w \times 3$

Image segmentation: for every pixel of an image, find the object it shows. Our example has two objects, a dog and the background.

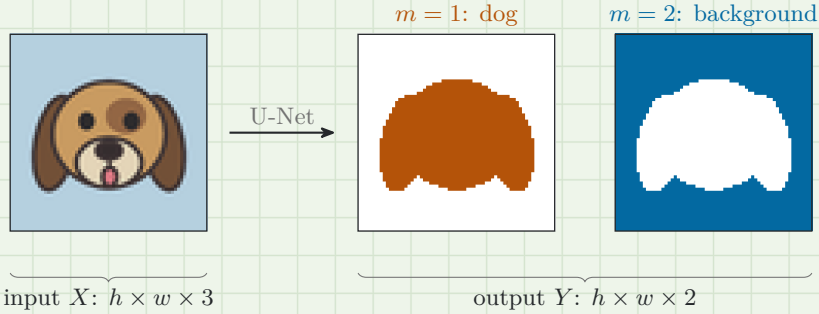
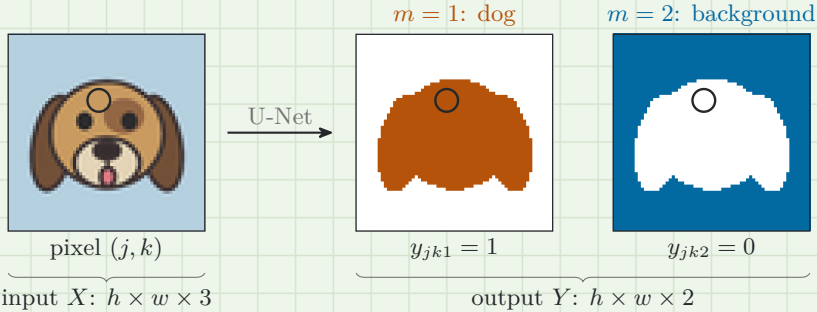


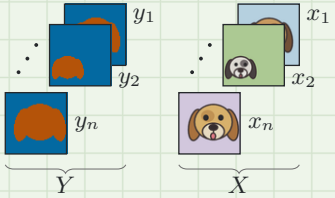
Image segmentation: for every pixel of an image, find the object it shows. Our example has two objects, a dog and the background.



$y_{jkm} = 1$ if pixel (j, k) shows object m, and $y_{jkm} = 0$ otherwise.

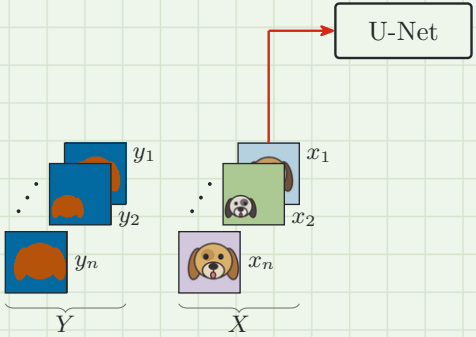
Method: the U-Net guesses the map of each image

The U-Net turns each image x_i into $\hat{y}_i$, a guess of its map y_i. The loss ℓ_i adds up the squared differences over every pixel (j, k) and every channel m.

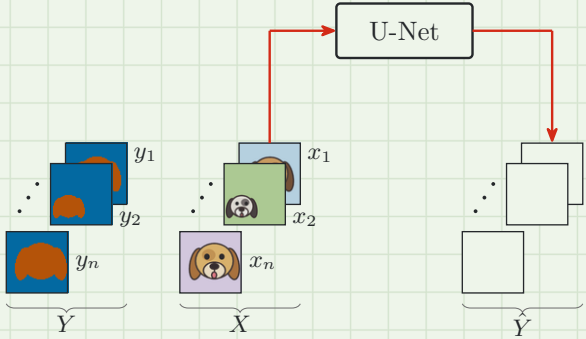


Method: the U-Net guesses the map of each image

The U-Net turns each image x_i into $\hat{y}_i$, a guess of its map y_i. The loss ℓ_i adds up the squared differences over every pixel (j, k) and every channel m.

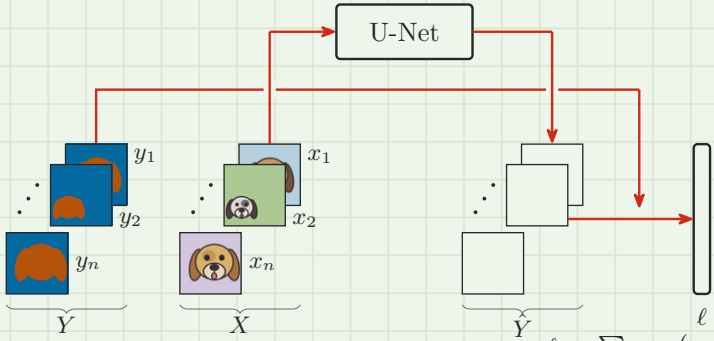


The U-Net turns each image x_i into $\hat{y}_i$, a guess of its map y_i. The loss ℓ_i adds up the squared differences over every pixel (j, k) and every channel m.



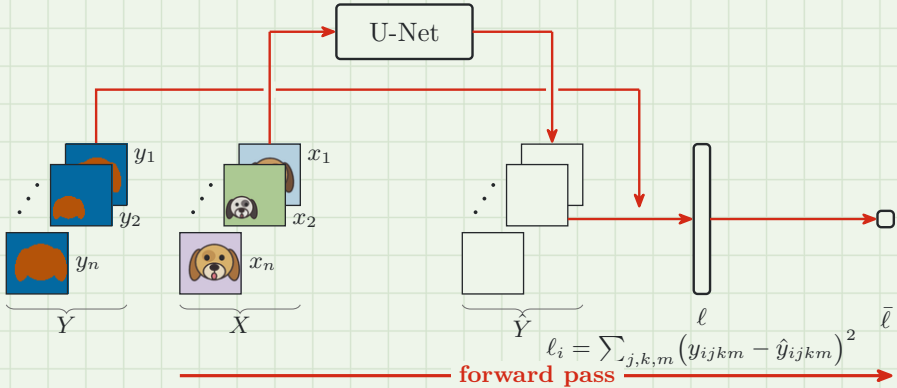
Method: the U-Net guesses the map of each image

The U-Net turns each image x_i into $\hat{y}_i$, a guess of its map y_i. The loss ℓ_i adds up the squared differences over every pixel (j, k) and every channel m.

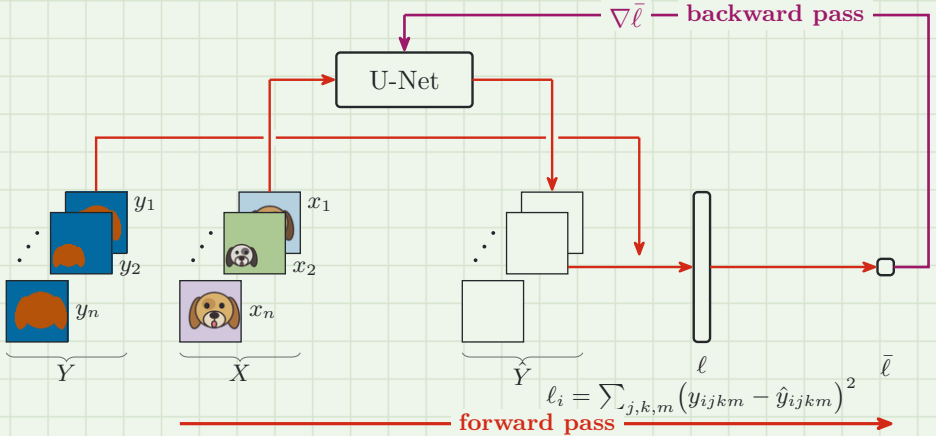


$$\ell_i = \sum_{j,k,m} \left(y_{ijkm} - \hat{y}_{ijkm}\right)^2$$

The U-Net turns each image x_i into $\hat{y}_i$, a guess of its map y_i. The loss ℓ_i adds up the squared differences over every pixel (j, k) and every channel m.



The U-Net turns each image x_i into $\hat{y}_i$, a guess of its map y_i. The loss ℓ_i adds up the squared differences over every pixel (j, k) and every channel m.



- A typical architecture for image segmentation.
- One can also treat a U-net as some sort of image autoencoder
- Image from original paper: https://arxiv.org/abs/1505.04597

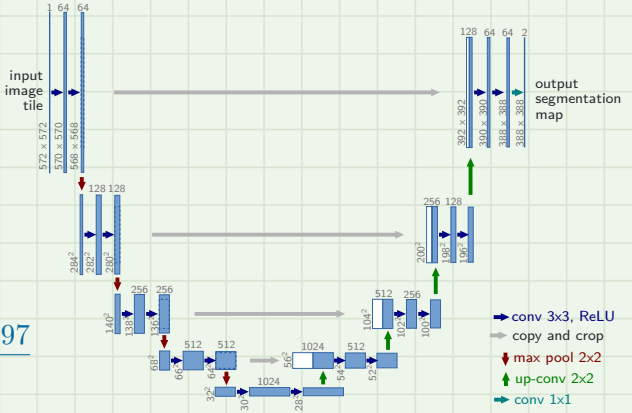


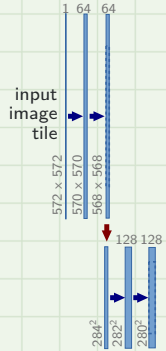
Fig. 1. U-net architecture (example for 32x32 pixels in the lowest resolution). Each blue box corresponds to a multi-channel feature map. The number of channels is denoted on top of the box. The x-y-size is provided at the lower left edge of the box. White boxes represent copied feature maps. The arrows denote the different operations.

After Fig. 1 of Ronneberger, Fischer and Brox, MICCAI 2015



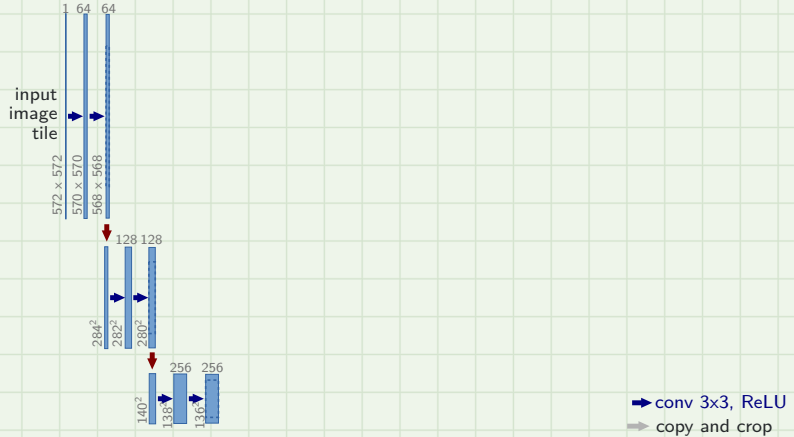
conv 3x3, ReLU
copy and crop
max pool 2x2
up-conv 2x2
conv 1x1

After Fig. 1 of Ronneberger, Fischer and Brox, MICCAI 2015

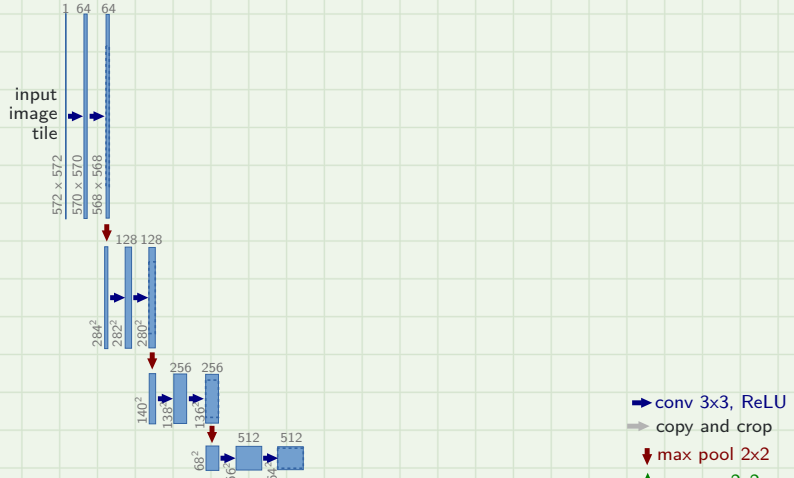


conv 3x3, ReLU
copy and crop
max pool 2x2
up-conv 2x2
conv 1x1

After Fig. 1 of Ronneberger, Fischer and Brox, MICCAI 2015



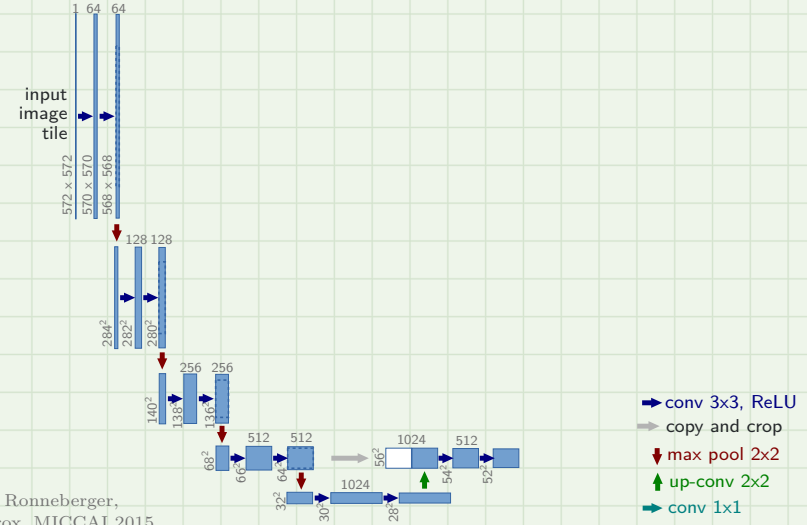
After Fig. 1 of Ronneberger, Fischer and Brox, MICCAI 2015



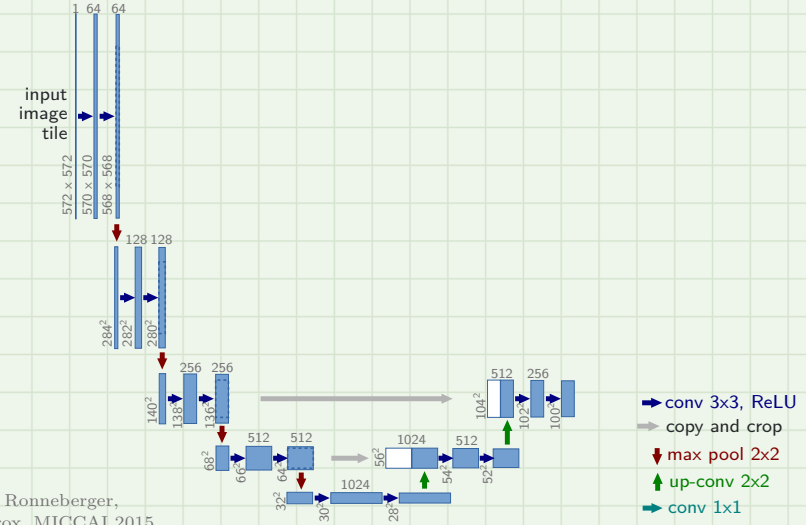
After Fig. 1 of Ronneberger, Fischer and Brox, MICCAI 2015



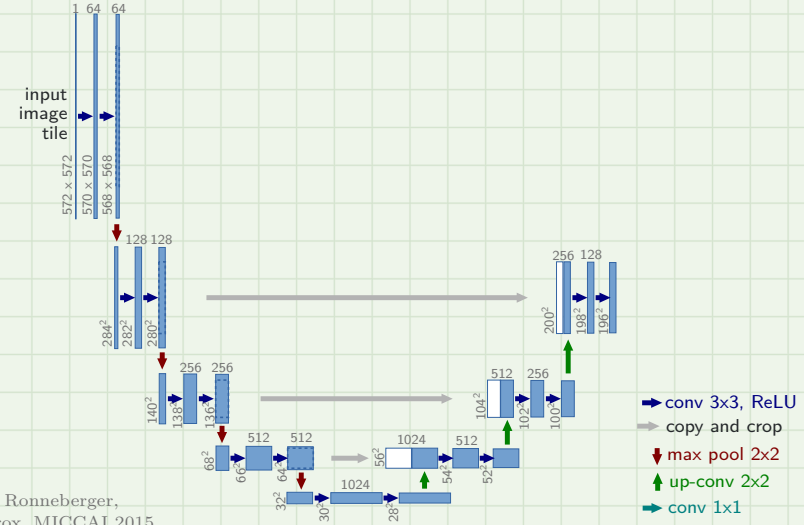
After Fig. 1 of Ronneberger, Fischer and Brox, MICCAI 2015



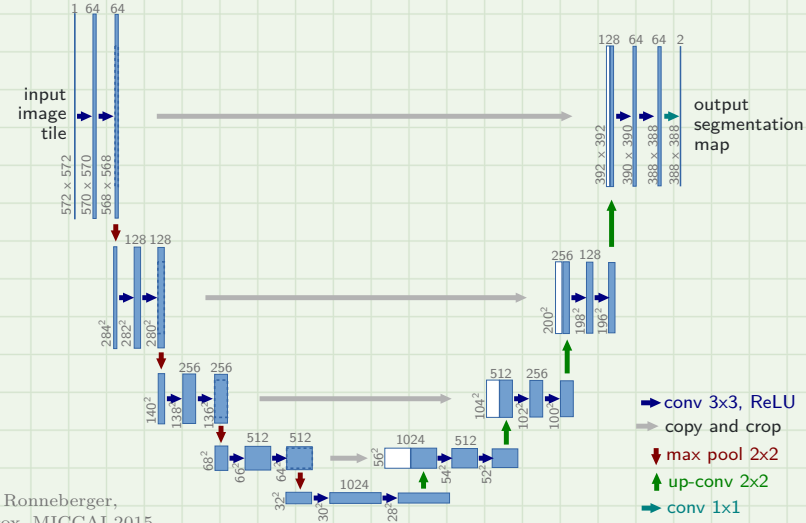
After Fig. 1 of Ronneberger, Fischer and Brox, MICCAI 2015



After Fig. 1 of Ronneberger, Fischer and Brox, MICCAI 2015



After Fig. 1 of Ronneberger, Fischer and Brox, MICCAI 2015



After Fig. 1 of Ronneberger, Fischer and Brox, MICCAI 2015